

MBI801: Information Systems Analysis and Design

Project theme: Pharmacy management system

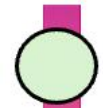


CHRISTCHURCH
CAMPUS

Group 5: *Claire Li; Abirami; Bryan Ferdinandus; Umanga Joshi*

Tutor: *Saranya Selvarangan*

Project Overview & System Context



System Background

- Many community pharmacies still rely on **manual prescription and inventory processes**, leading to
 - low efficiency
 - data errors and delays
 - poor customer experience



Proposing: system provides an integrated online pharmacy management platform to connect customers, pharmacists and doctors.

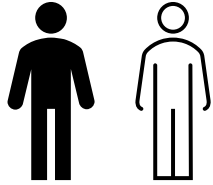


System Objectives

- Enable **secure online prescription upload and approval**.
- Support **real-time inventory tracking and automatic replenishment**.
- Provide **order and payment management** with instant notifications.
- Improve **accuracy, data integrity and customer satisfaction**.



Use Cases Summary



Customer

Process order
Manage customer profile
Upload prescription

Upload prescription

- Validation
- Secure storage
- Notification

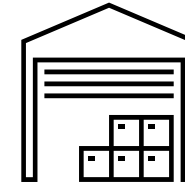


Pharmacist

Review prescription
Notify customer
Prepare medication

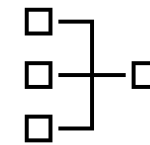
Preparing Medication

- Medication preparation
- Customer communication
- Stock management



Supplier

Update inventory



System

Check stock
Trigger recorder
Issue receipt

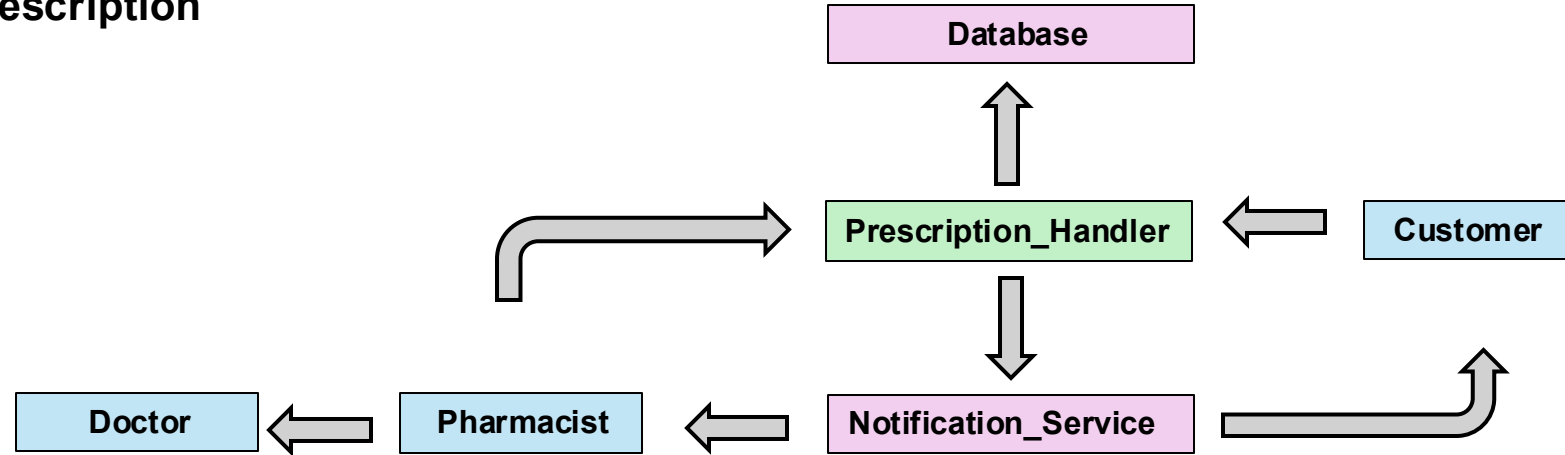
CRC Cards - Both Use Cases

Class

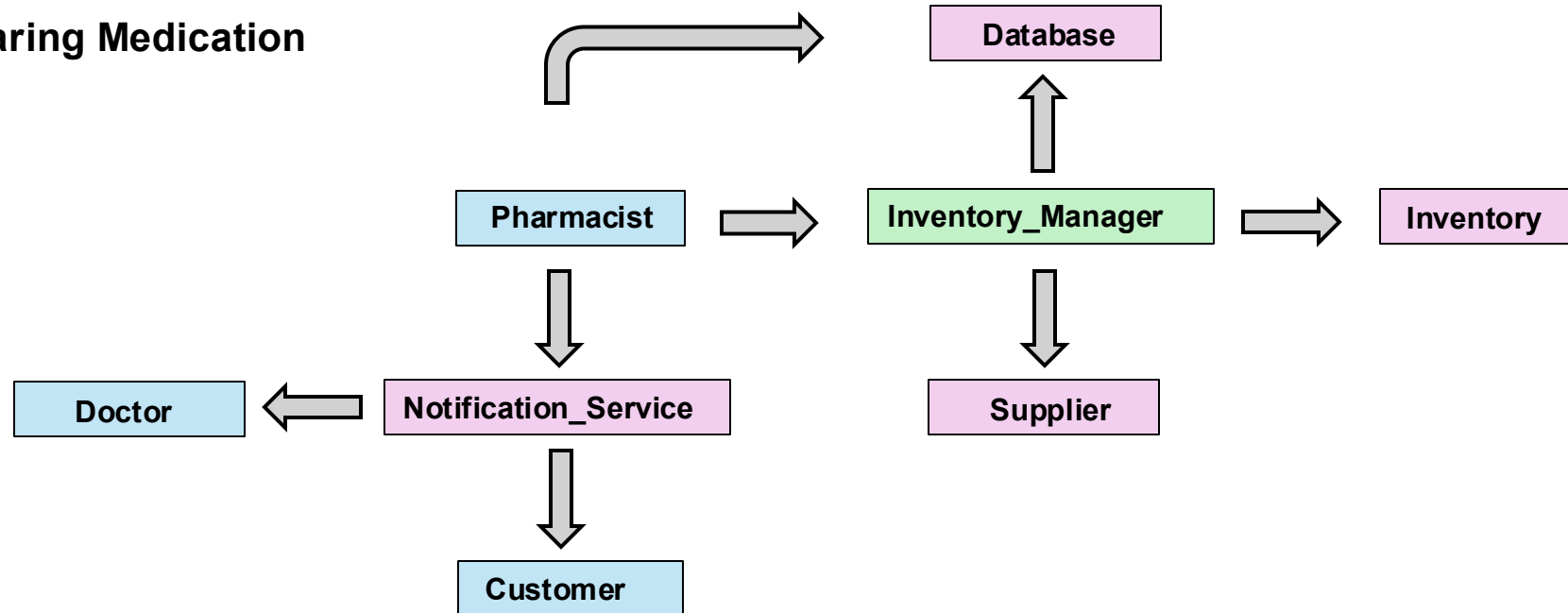
Responsibility

Collaborator

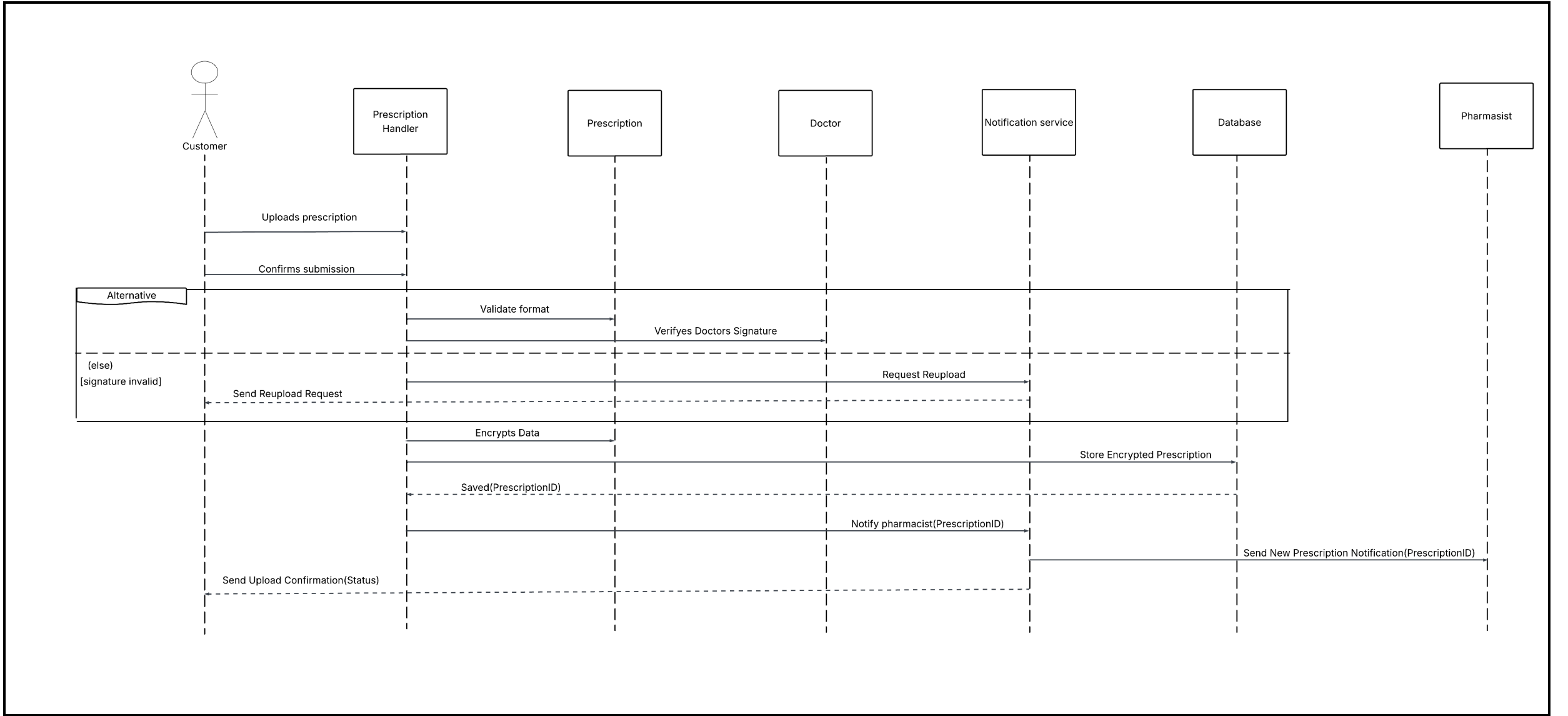
Upload Prescription



Preparing Medication

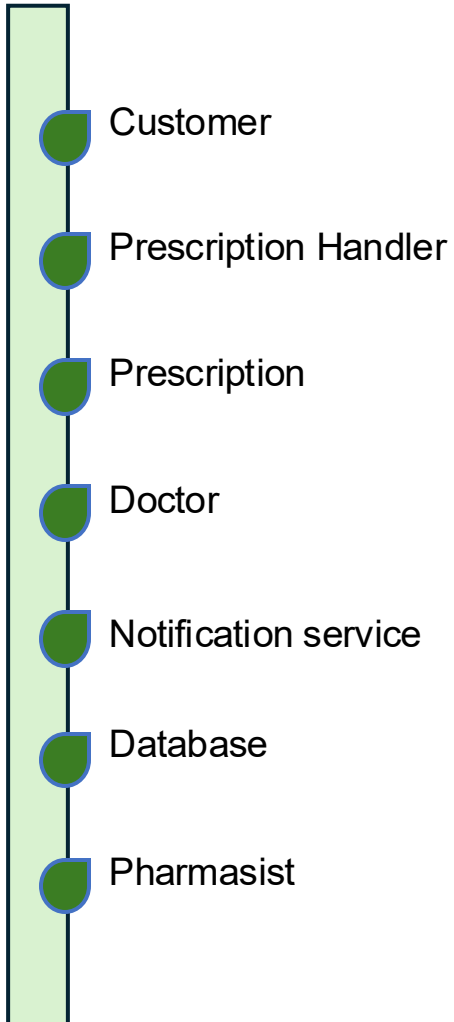


Sequence Diagram - Use Case 1(Upload Prescription)



Sequence Diagram - Use Case 1

Upload Prescription



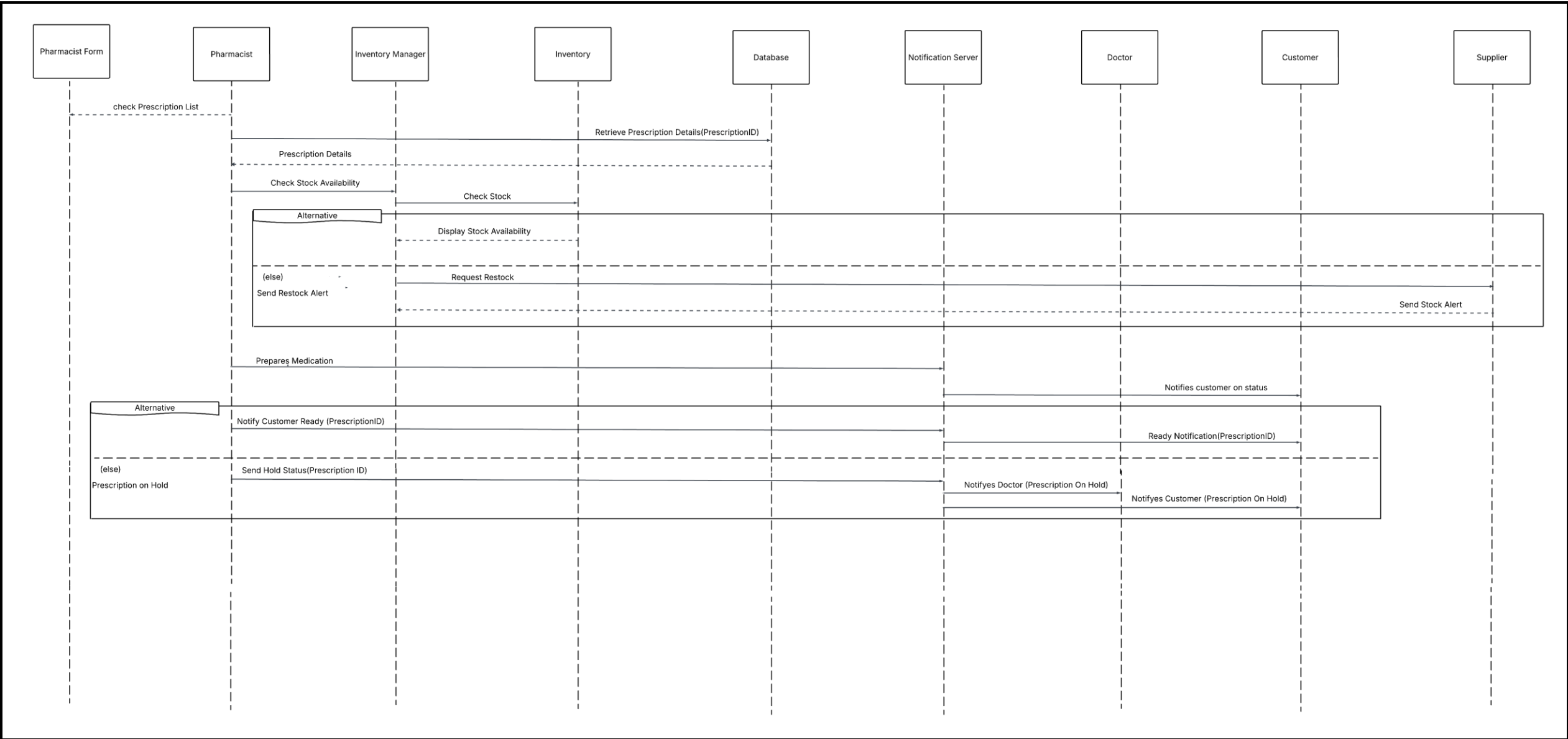
Key Steps

- The customer upload and confirms the prescription
- Prescription Handler validates the prescription and verifies the doctor signature
- Alternative – If the doctor signature is invalid → Reupload request sent to the customer
- If signature valid → Data encrypted and stored in Database
- Notification service alters the Pharmacist of a new prescription
- Customer receives upload confirmation

Highlight

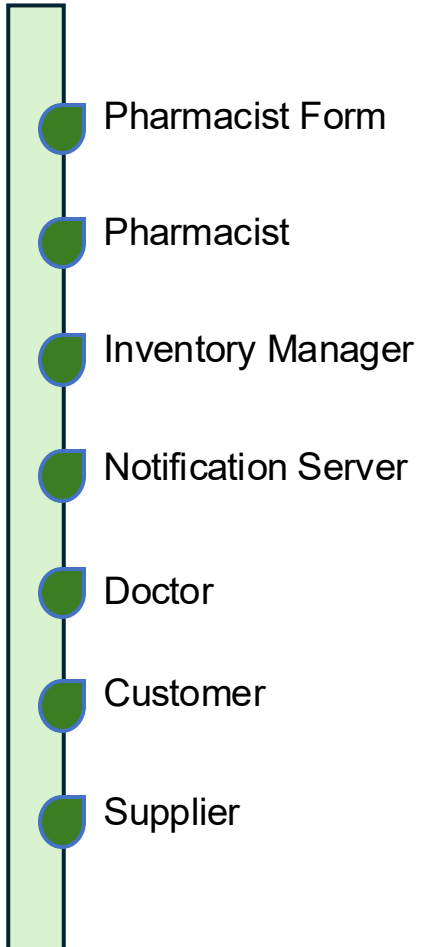
- Automatic doctor verification and pharmacist notification
- Secure data encryption and storage
- Alternative method input for invalid format of prescription
- Reliable workflow automation

Sequence Diagram - Use Case 2 (Preparing Medication)



Sequence Diagram - Use Case 2

Preparing Medication



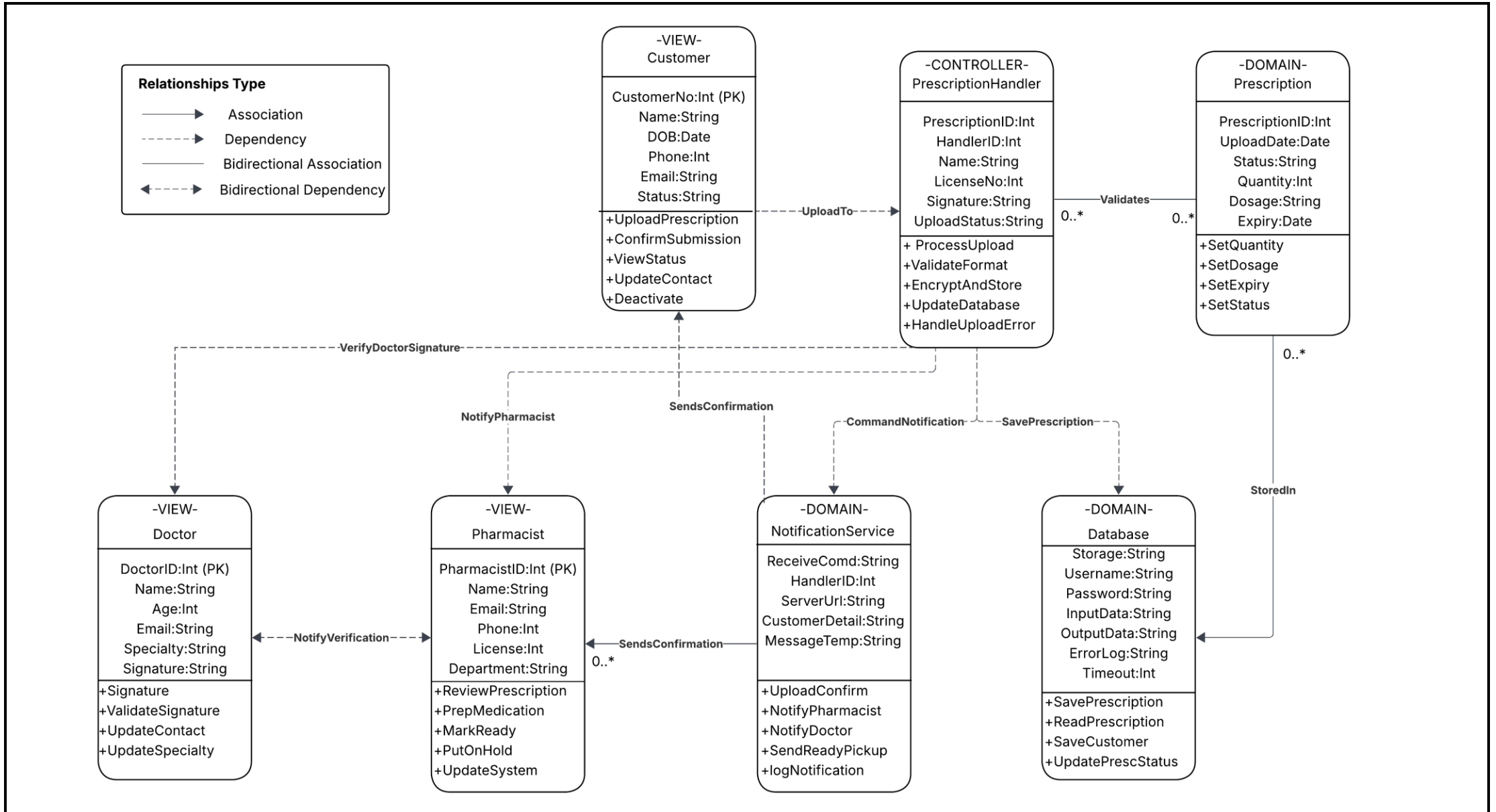
Key Steps:

- Pharmacist form request prescription list of the customer
- Pharmacist fetch the prescription data from the database
- Inventory manager checks on the stock availability
- If the stock not available → restock alert is sent to the supplier and customer is informed of delay.
- If the stock is available → the pharmacist starts to prepare the medication
- Once the prescription is prepared , Ready Notification is sent to the customer
- If not ready , Hold status notification is sent to both Doctor and Customer

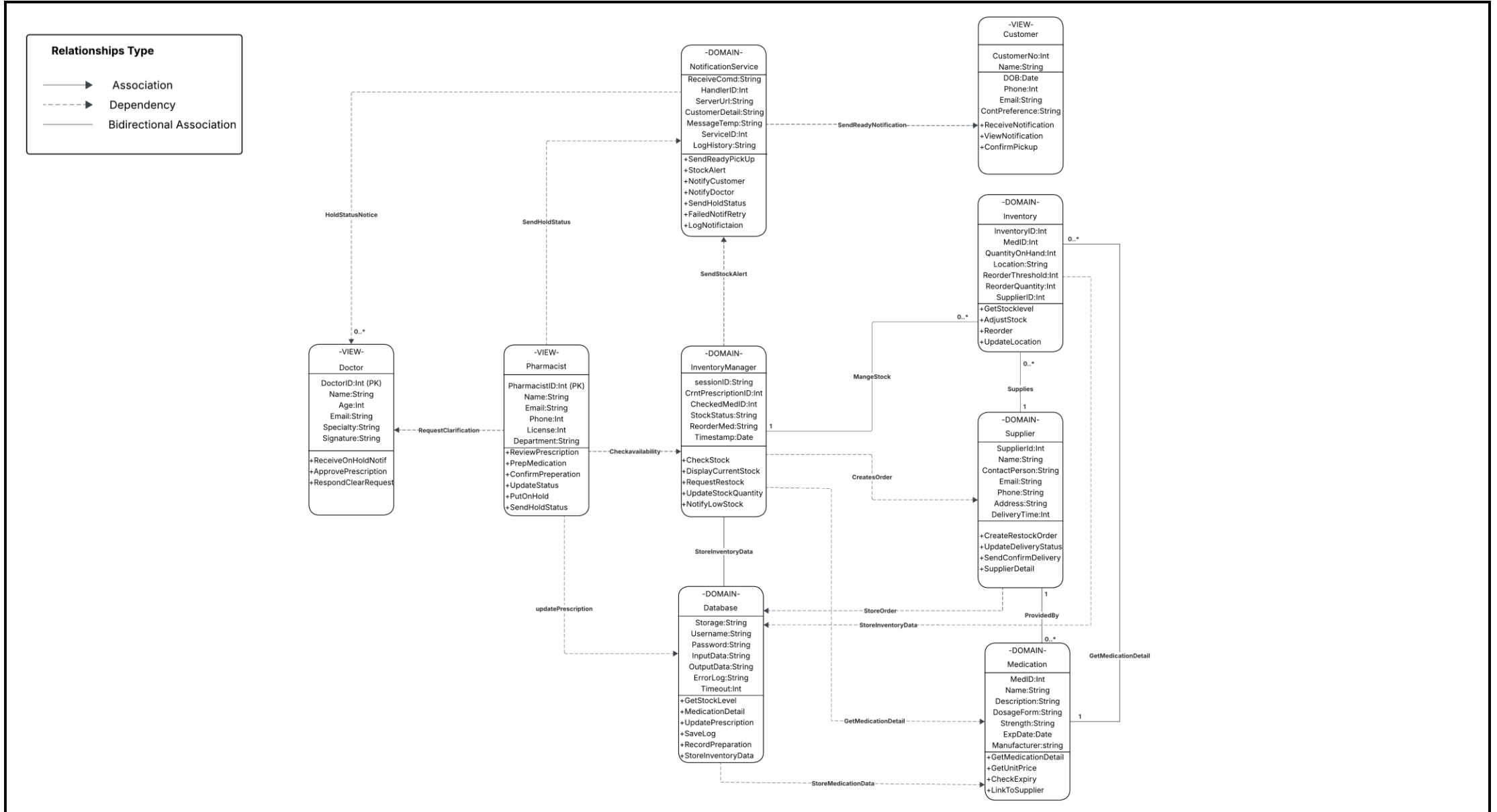
Highlights:

- Well organised automated restock alert sent to the supplier
- Notifying customer and doctor improve transparency on order status
- Automated stock verification to ensure medicine availability

Design Class Diagram- Use Case 1(Upload Prescription)



Design Class Diagram - Use Case 2 (Preparing Medication)



Risk analysis



Scalability

- Performance Bottlenecks
- Inadequate Infrastructure
- Ineffective Database Structure
- Integration Limitation

Data integrity

- Human Error
- System Failures or Crashes
- Unauthorised Data Access
- Data Synchronisation



Cultural Perspectives Integration



Te Ao Maori

- whanaungatanga (relationships)
- manaakitanga (care and respect)
- kaitiakitanga (guardianship)

User Needs and Requirements

- **Secure prescription uploads**
- **Automatic inventory updates**
- **Real-time order tracking**



Conclusions and Key Takeaways

Conclusions

- Efficiently automates prescription handling ,stock monitoring and notifications.
- Ensures data security, scalability and smooth coordination between Pharmacist , customer and supplier.
- The design also reflects cultural inclusiveness and ethical responsibility.

Key Takeaways

- Usage of structured modelling (CRC, Sequence, Design Class Diagram)
- Automated validation and stock alert which shows workflow efficiency
- Reliable encryption and data integrity practise
- User friendly and culturally responsive interface

References

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Thanks !!!

